ANNA G. MORGANI (DEVLEN)

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PROFESSIONAL SUMMARY

Organized research scientist with three years of industry experience and an MS in Molecular Science and Software Engineering from UC Berkeley. Creative problem-solver with demonstrated experience bridging the gap between wet-lab assay development and computational modeling to accelerate therapeutic development and enhance patient outcomes. Seeking opportunities that challenge me to apply my software engineering skills to new problems in biology.

EDUCATION

University of California, Berkeley | Berkeley, CA **May 2026**
Master of Science in Molecular Science and Software Engineering, GPA: 3.82
Relevant Coursework: Software Engineering for Scientific Computing, Machine Learning Algorithms

Rensselaer Polytechnic Institute | Troy, NY **May 2022**
Bachelor of Science in Biochemistry/Biophysics, GPA: 3.58
Leadership: Board Member, Women at Rensselaer Mentoring; Executive Administrator, Alpha Phi (Theta Tau)

SKILLS

Programming Languages: Python, C++, R, SQL, Rust
Software Engineering: Version Control (Git), Cloud Computing (Azure), Docker, Object-Oriented Programming
Machine Learning: Neural Networks (ANN/GAT), PyTorch, TensorFlow, Scikit-learn, Gradient Boosting Machines, Support Vector Machines, Linear & Logistic Regression
HPC & Parallel Computing: MPI, OpenMP, Strong Scaling Analysis, Load Balancing, Slurm
AI & Developer Productivity: Cursor, GitHub Copilot, Prompt Engineering
Wet Lab: Multiplex Flow Cytometry, Cell-based Assays, Immunoassays (Luminex, MSD, ELISA), qPCR

PROFESSIONAL EXPERIENCE

Lucy Therapeutics | Waltham, MA **November 2024 - May 2025**
Senior Research Associate
Biotechnology company developing molecules targeting mitochondria to treat neurodegenerative diseases.
- Developed an F-ATPase hydrolase assay with purified protein to assess mechanism of action and competition, automating the characterization of protein-protein competition using R synergyfinder and medianeffect packages
- Supported data package generation for 50+ Phase 0 clinical trial samples through isolation of patient mitochondrial DNA and qPCR to assess mitochondrial DNA damage
- Facilitated weekly technical meetings to a cross-functional team, delivering data-driven recommendations on assay performance and troubleshooting that informed internal drug discovery milestones

Bicycle Therapeutics | Cambridge, MA **May 2022 - August 2024**
Associate Research Scientist, Immuno-Oncology
Biopharmaceutical company specializing in phage-display for creation of peptides for cancer immunotherapy.
- Implemented an R script to automate analysis of dose-response relationships from a multiplex cytokine readout, accelerating program progress by reducing data processing time by 40%
- Generated quantitative flow cytometry data characterizing human PBMCs, directly influencing recommended phase II dose modeling for a clinical candidate
- Supported the on-time submission of an IND application through generation of quantitative flow cytometry data, creation of summary figures, and rigorous review of in vitro pharmacology data
- Authored and standardized SOPs for laboratory protocols and data analysis workflows; used these materials to train a co-op student, enabling independent execution of complex assays and maximizing knowledge transfer
- Strategically communicated complex cell-based assay results to diverse scientific audiences, serving as a primary point of contact for internal stakeholders to facilitate data-driven decision-making for clinical candidates

HIGHLIGHTED PROJECTS

ML Algorithms to Decode Cancer Metastasis | UC Berkeley & Genialis **January 2026 - Present**

- Collaborated with industry sponsor Genialis and a small team of graduate students to implement Azure-based machine learning algorithms trained on single cell RNA-seq data to model cancer metastasis pathways
- Managed project timelines, facilitated cross-functional technical syncs, and tracked project progress through weekly review of meeting minutes as the Docs & DevOps lead
- Facilitated weekly technical presentations to the Genialis team, iteratively refining model parameters based on stakeholder feedback

Neurodegenerative Disease Classification | UC Berkeley **October 2025 - December 2025**

- Co-designed a PyTorch-based Artificial Neural Network (ANN) and Graph Attention Network (GAT) to predict neurodegenerative disease status from proteomics data, achieving 87% accuracy on unseen data with the ANN architecture
- Acted as the Biology SME for a cross-functional technical team, translating complex signaling pathways and protein-protein interaction (PPI) networks into actionable features for model architecture
- Leveraged industry experience in neurodegeneration to guide the selection of biological markers, ensuring the computational model's outputs remained biologically relevant

Spectral Clustering for Single-Cell RNA Sequencing | UC Berkeley **October 2025 - December 2025**

- Co-implemented a spectral clustering algorithm from scratch to categorize immune cell populations within a single cell RNA-seq dataset, obtaining clusters that aligned with the scikit-learn SpectralClustering algorithm
- Provided domain expertise in immunology to guide feature selection and validate clustering results against known biological markers

SELECTED PUBLICATIONS & PRESENTATIONS

Lahdenranta, J., Hurov, K., … **Devlen, A.**, et al. (2024). *Tumor-targeted activation of CD137 using Bicycle® molecules : New insights into mechanism of action and discovery of BT7455, a clinical candidate for the treatment of EphA2-expressing cancers*. [Abstract] Proceedings of the AACR Annual Meeting; Abstract nr 5301.
***Key Contribution:** Conducted multiplex flow cytometry characterization of human PBMCs and cancer cell lines (A549, PC3) to validate phenotypic markers and ensure high-fidelity biological inputs for clinical candidate characterization.

Luus, L., Hurov, K., … **Devlen, A.**, et al. (2023). *EphA2-dependent CD137 agonism and anti-tumor efficacy by BT7455, a Bicycle tumor-targeted immune cell agonist®*. [Abstract] Proceedings of the AACR Annual Meeting; Abstract nr 1832.
***Key Contribution:** Standardized multiplex flow cytometry protocols for MC38 and other cancer cell lines to minimize biological variability and ensure experimental reproducibility across efficacy studies.